Digital Image Processing

Lecture 05: Image Segmentation

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Readings: Chapter 10 (sections 10.1-10.3)
How to read: MATH alert! read and experiment!

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Fundamentals of Image Segmentation

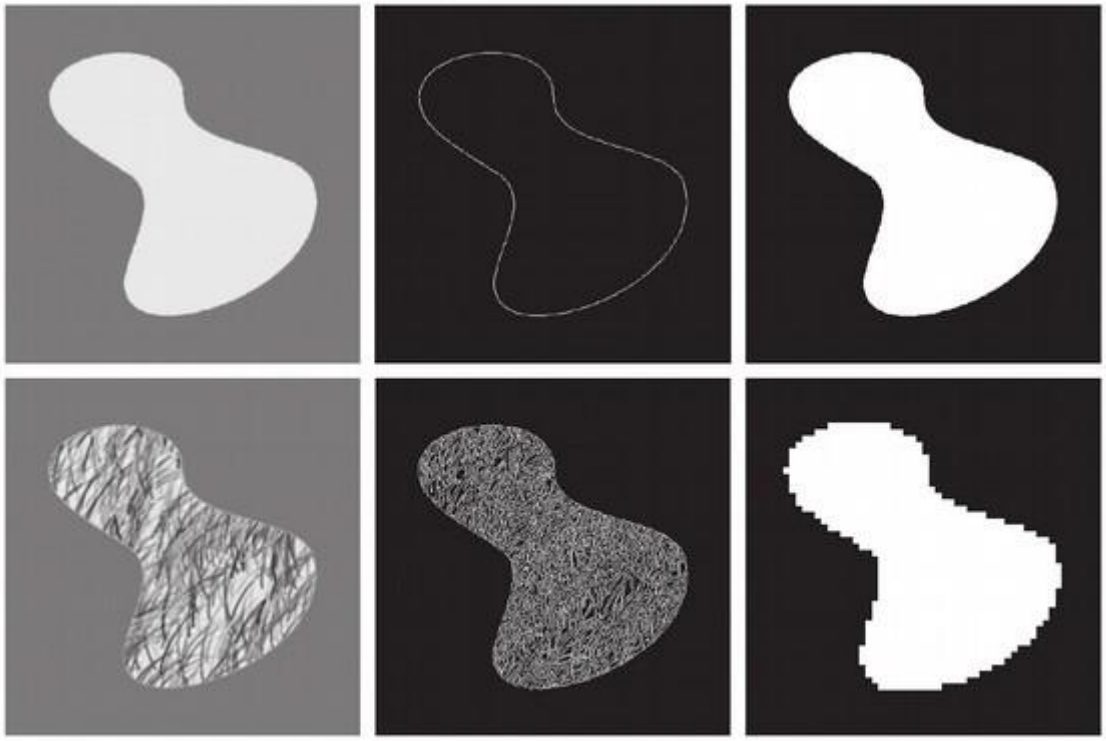
- Assume **I** is the entire spatial region of an image. Image segmentation is the process of dividing **I** into partitions $\mathbf{R_1}$, $\mathbf{R_2}$, …, $\mathbf{R_n}$ such that:

  1) Union: $\bigcup_{i=1}^{n} R_i = I$
  2) Connectivity : $\mathbf{R_i}$ is a connected set for **i=1,2,3,...,n**
  3) Disjoint: $R_i \cap R_j = \varnothing$ for $i \neq j$
  4) Intra-region similarity: $\mathbf{Q(R_i)=TRUE}$ for **i=1,2,3,...,n**
  5) Inter-regions dissimilarity: $\mathbf{Q(R_i \cup R_j) = FLASE}$ for $i \neq j$

- Here $\mathbf{Q(R_k)}$ is a logical predicate defined over the points in set $\mathbf{R_k}$.

Fundamentals of Image Segmentation

- Segmentation algorithms are generally based on **two basic** categories:
  - ***Discontinuity***: boundaries of regions are sufficiently different from each other and from background → **Edge-based** segmentation
  - ***Similarity***: partitioning an image into regions that are similar according to some criteria → **Region-based** segmentation.

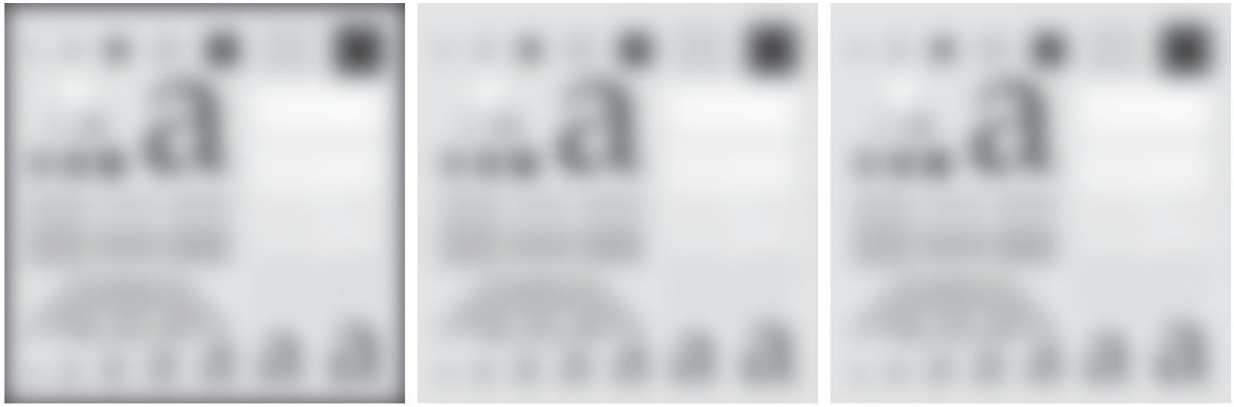


a b c
d e f

FIGURE 10.1 (a) Image of a constant intensity region. (b) Boundary based on intensity discontinuities. (c) Result of segmentation. (d) Image of a texture region. (e) Result of intensity discontinuity computations (note the large number of small edges). (f) Result of segmentation based on region properties.

Image Padding?

- We saw image zero-padding when computing correlations and convolutions.
- Padding methods:
  - Zero-padding
  - Mirror (or symmetric) padding: values outside the boundary are obtained by mirror-reflecting the image across the border.
  - Replicate padding: values outside the boundary are set equal to the nearest image border value.



a b c

FIGURE 3.39 Result of filtering the test pattern in Fig. 3.36(a) using (a) zero padding, (b) mirror padding, and (c) replicate padding. A Gaussian kernel of size 187×187, with $K = 1$ and $\sigma = 31$ was used in all three cases.

High-Pass and Low-Pass Filtering in Image Processing

Low-Pass Filtering (LPF)

- A linear filtering technique that **preserves low spatial frequency components** while attenuating high-frequency components
- Primarily used to **reduce noise and smooth images**
- Implemented via **convolution** with smoothing kernels (e.g., mean, Gaussian)
- Effect: **Blurring and reduction of fine details**

High-Pass Filtering (HPF)

- A filtering technique that **preserves high spatial frequency components** while suppressing low-frequency components
- Used to **enhance edges and fine details**
- Often implemented using **derivative-based operators** (e.g., Sobel, Laplacian)
- Effect: **Edge enhancement and image sharpening**

High-Pass and Low-Pass Filtering in Image Processing

Low-Pass Filter (Example Kernel)

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- Performs local averaging
- Reduces intensity variations and noise

High-Pass Filter (Example: Laplacian Operator)

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

- Approximates second-order spatial derivatives
- Highlights regions with rapid intensity change

Finite Differences

- We saw that *local averaging* smooths an image. Also we saw *intensity variations* can be detected by first and second order derivatives.
- Generally speaking, derivatives in images are done in terms of finite differences.
- First-order derivative:
  - Must be ***zero*** in areas of **constant** intensity;
  - Must be ***non-zero*** at the onset of an intensity **step** or **ramp**;
  - Must be ***non-zero*** along intensity **ramps**.
- Second-order derivative:
  - Must be ***zero*** in areas of **constant** intensity;
  - Must be ***non-zero*** at the onset and end of an intensity **step** or **ramp**;
  - Must be ***zero*** along intensity **ramps**.

Sharpening (Highpass) Spatial Filters

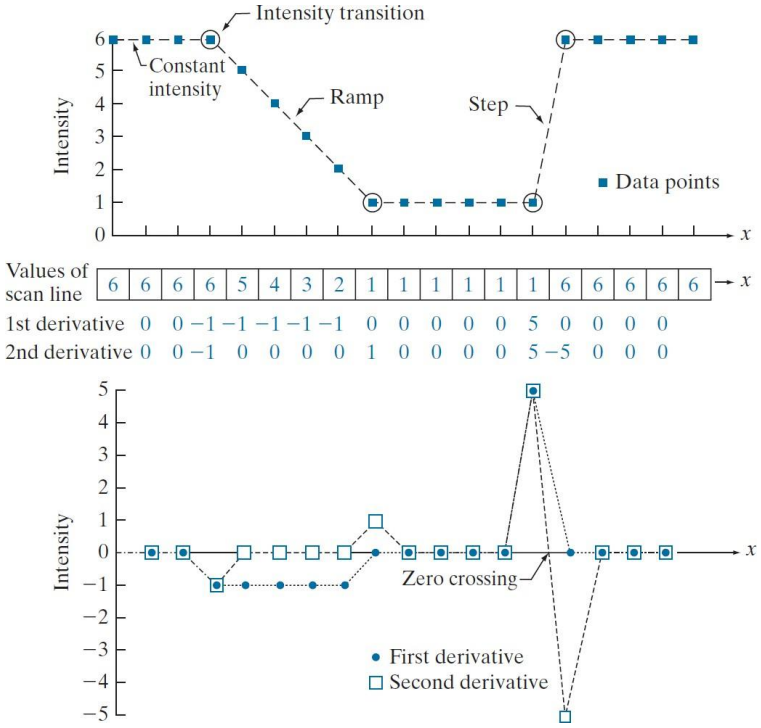
- For digital functions, the derivatives are defined in terms of differences.
- **First-order** derivative of a 1D function in difference form:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- **Second-order** derivative of a 1D function in difference form:

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

- Partial derivative is used, so the notation is consistent between 1D and 2D functions.

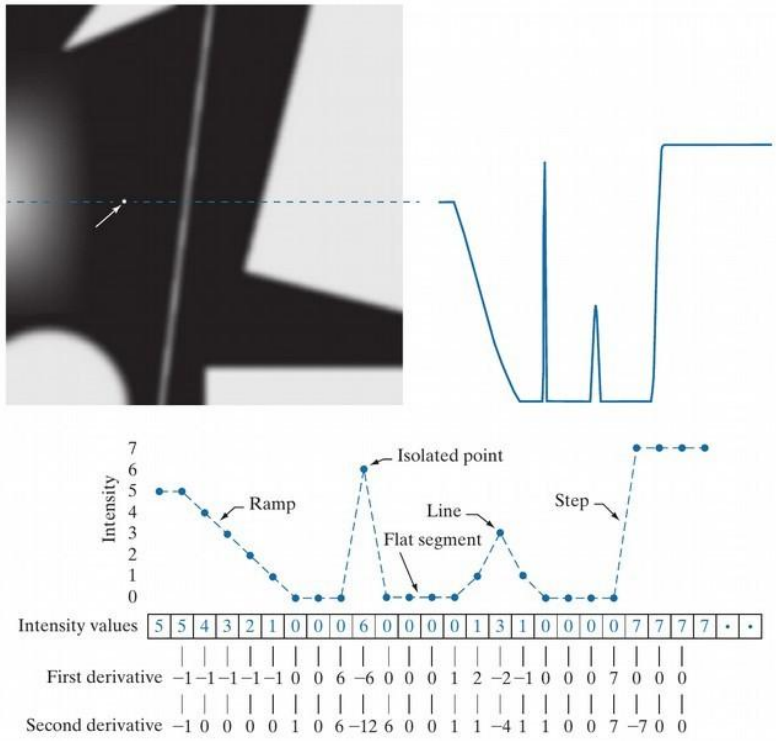


Finite Differences

- **First derivative** produces thick edges in images.
- **Second derivative** has a stronger response to fine details, such as thin lines, isolated points and noise.
- **Second derivative** produces double-edge response at ramp and step transitions, useful to locate edges.
- The sign of the **second derivative** can be used to see if a transition into an edge is from dark to light or light to dark.

a b
c

FIGURE 10.2 (a) Image. (b) Horizontal intensity profile that includes the isolated point indicated by the arrow. (c) Subsampled profile; the dashes were added for clarity. The numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10-4) for the first derivative and Eq. (10-7) for the second.



Point Detection

- As we saw, **second-order derivative** can be used for detecting isolated points.
- We have seen Laplacian before:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- Using a central difference formulation:

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

0	1	0
1	−4	1
0	1	0

Point Detection

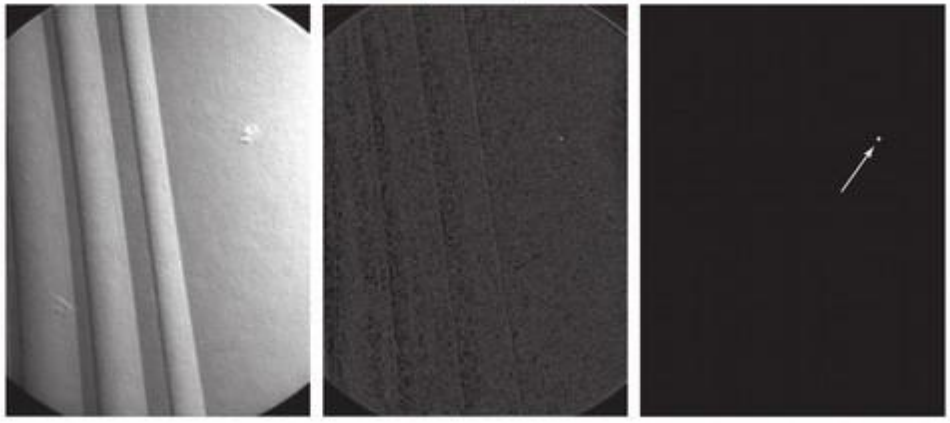
- Using the **Laplacian operator**:
  - Compute the Laplacian response at every pixel;
  - Take the absolute value of Laplacian response;
  - If the result is greater than a threshold, set the pixel to 1. Otherwise, set the pixel to 0.

$$g(x,y) = \begin{cases} 1 & |\nabla^2 f(x,y)| \geq T \\ 0 & otherwise \end{cases}$$

a
b c d

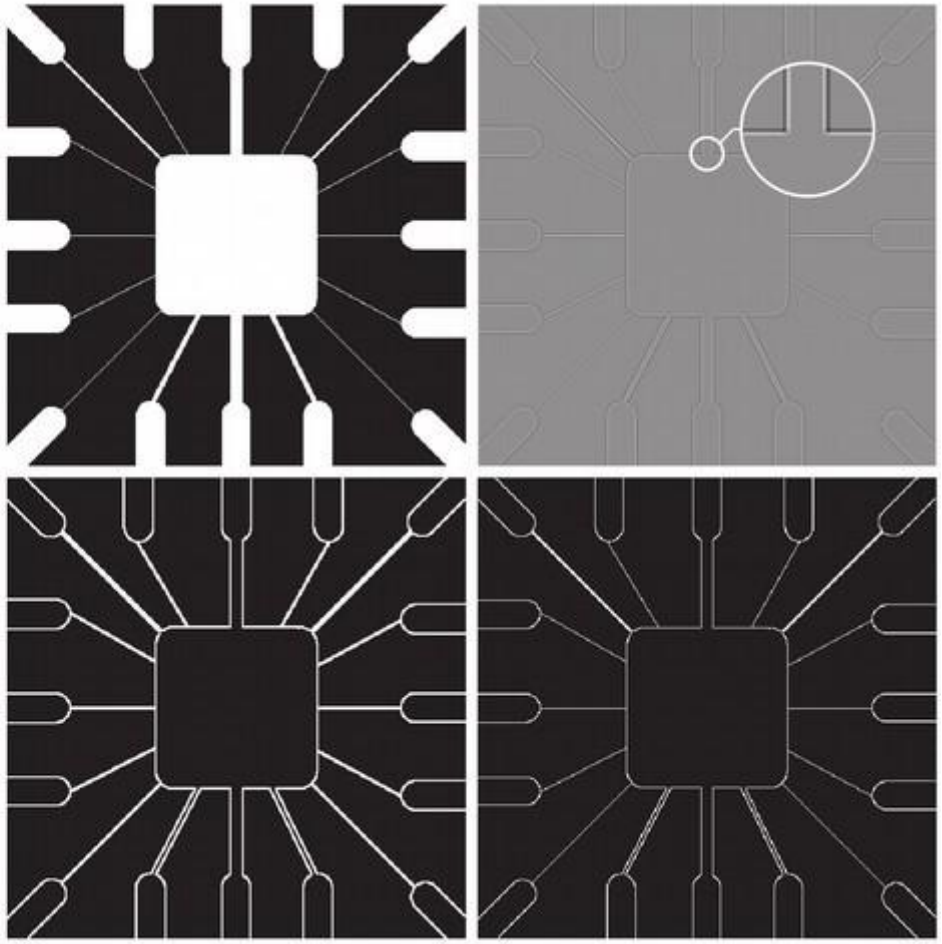
FIGURE 10.4 (a) Laplacian kernel used for point detection. (b) X-ray image of a turbine blade with a porosity manifested by a single black pixel. (c) Result of convolving the kernel with the image. (d) Result of using Eq. (10-15) was a single point (shown enlarged at the tip of the arrow). (Original image courtesy of X-TEK Systems, Ltd.)

1	1	1
1	−8	1
1	1	1



Line Detection

- Using the second-order derivative results in a stronger filter response and thinner lines in comparison to first-order derivatives.



a b
c d

FIGURE 10.5 (a) Original image. (b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian. (c) Absolute value of the Laplacian. (d) Positive values of the Laplacian.

Line Detection

- We can define filter kernels to detect lines in specified directions.

−1	−1	−1
2	2	2
−1	−1	−1

Horizontal

2	−1	−1
−1	2	−1
−1	−1	2

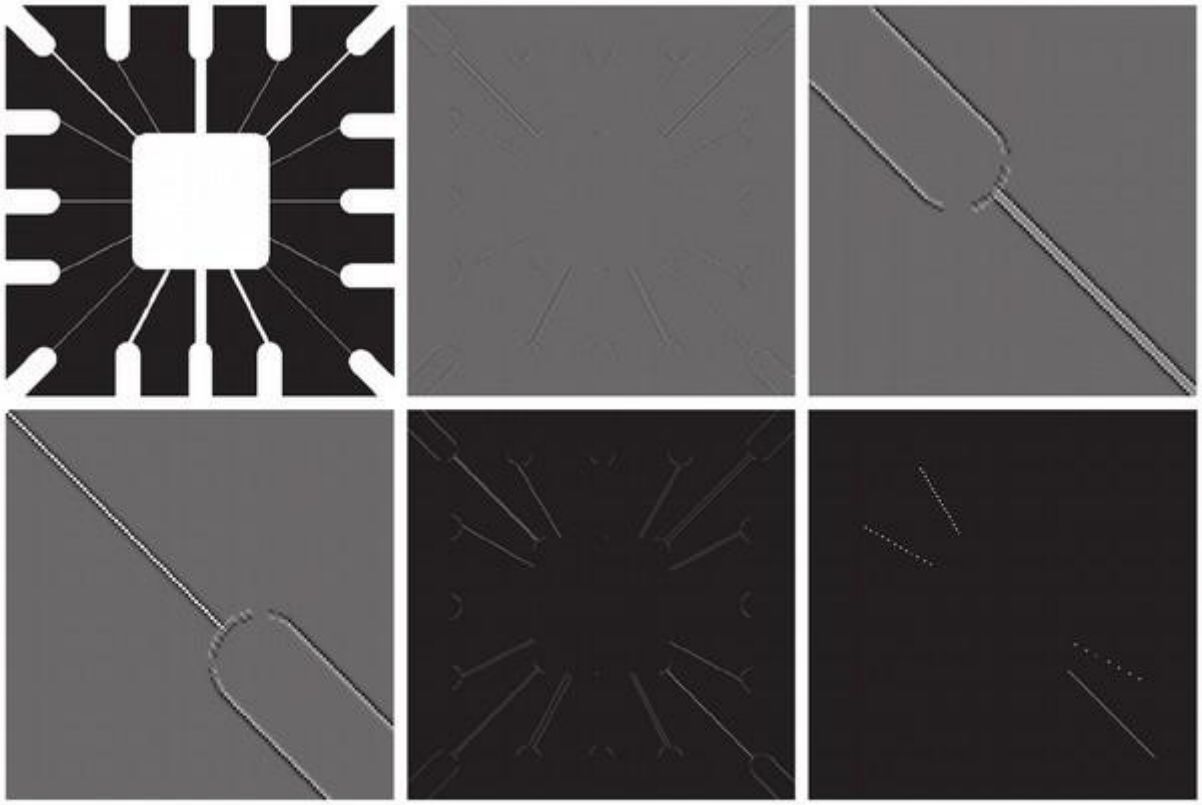
+45°

−1	2	−1
−1	2	−1
−1	2	−1

Vertical

−1	−1	2
−1	2	−1
2	−1	−1

−45°

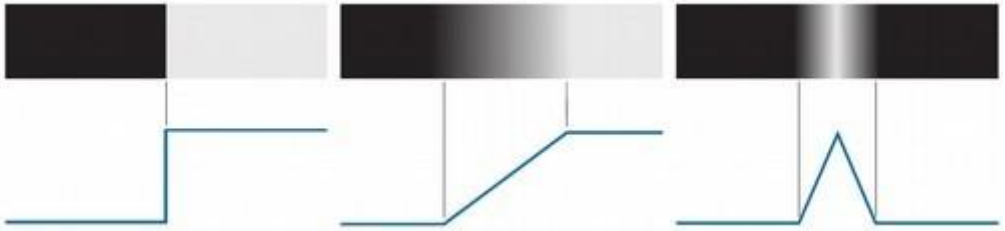


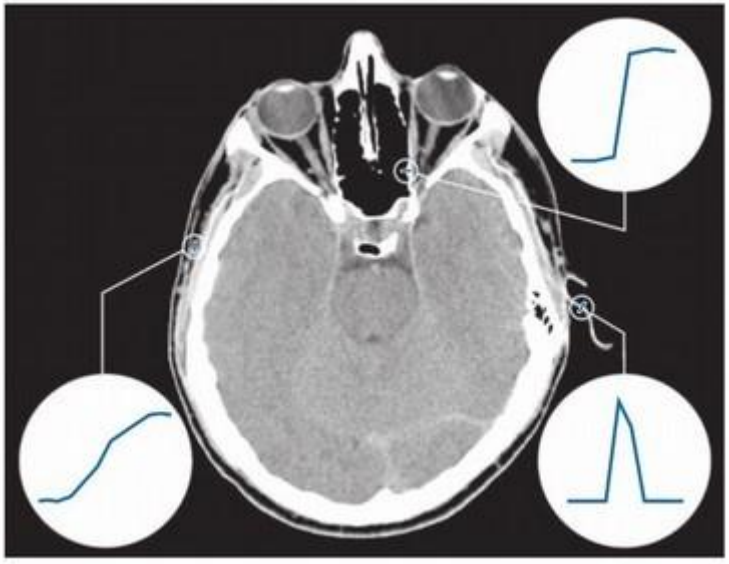
a b c
d e f

FIGURE 10.7 (a) Image of a wire-bond template. (b) Result of processing with the +45° line detector kernel in Fig. 10.6. (c) Zoomed view of the top left region of (b). (d) Zoomed view of the bottom right region of (b). (e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g > T$, where g is the image in (e) and $T = 254$ (the maximum pixel value in the image minus 1). (The points in (c) were enlarged to make them easier to see.)

Edge Detection

- Edge detection is a common approach for image segmentation based on **local intensity changes** and **discontinuities**.
- Three types of edges:
  - ***Step edge***: a transition between two intensity levels occurring ideally over the distance of one pixel;
  - ***Ramp edge***: a gradual transition between two intensity levels
  - ***Roof edge***: intensities on both sides of the roof are similar, with different intensity in the middle.



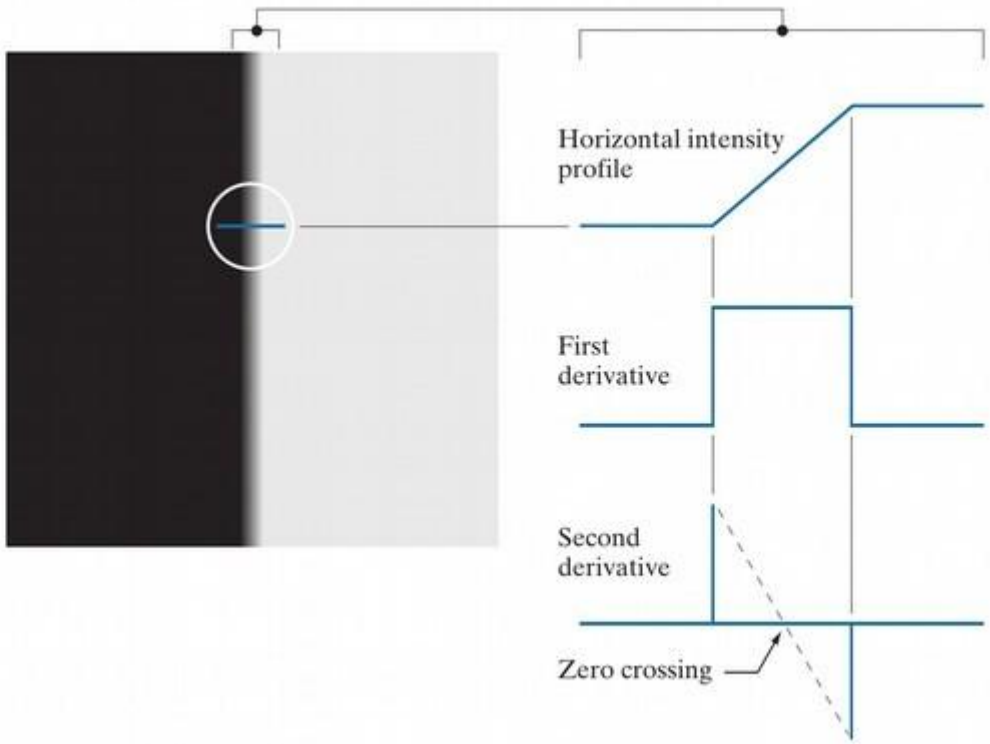


Edge Detection

- The first derivative produces a **thick edge** profile.
- The second derivative produces a d**ouble-edge** profile with a sign change; the point of zero-crossing can be used to locate the edge.

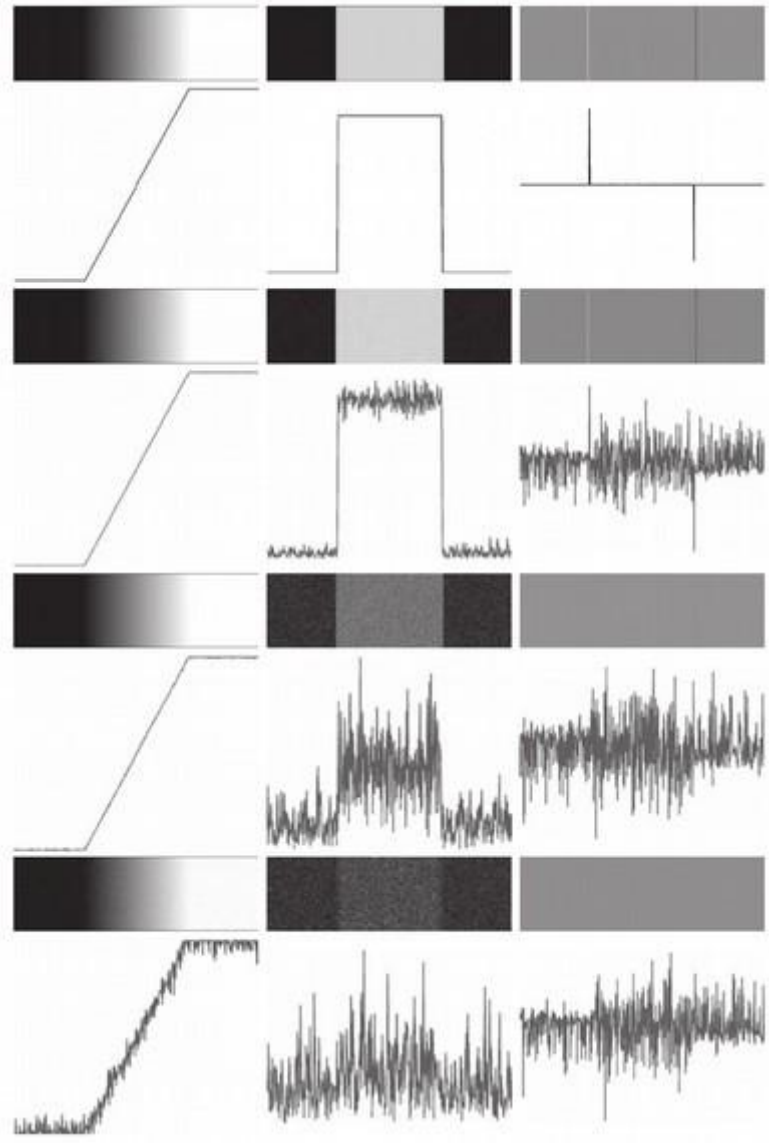
a b

FIGURE 10.10
(a) Two regions of constant intensity separated by an ideal ramp edge. (b) Detail near the edge, showing a horizontal intensity profile, and its first and second derivatives.



Edge Detection

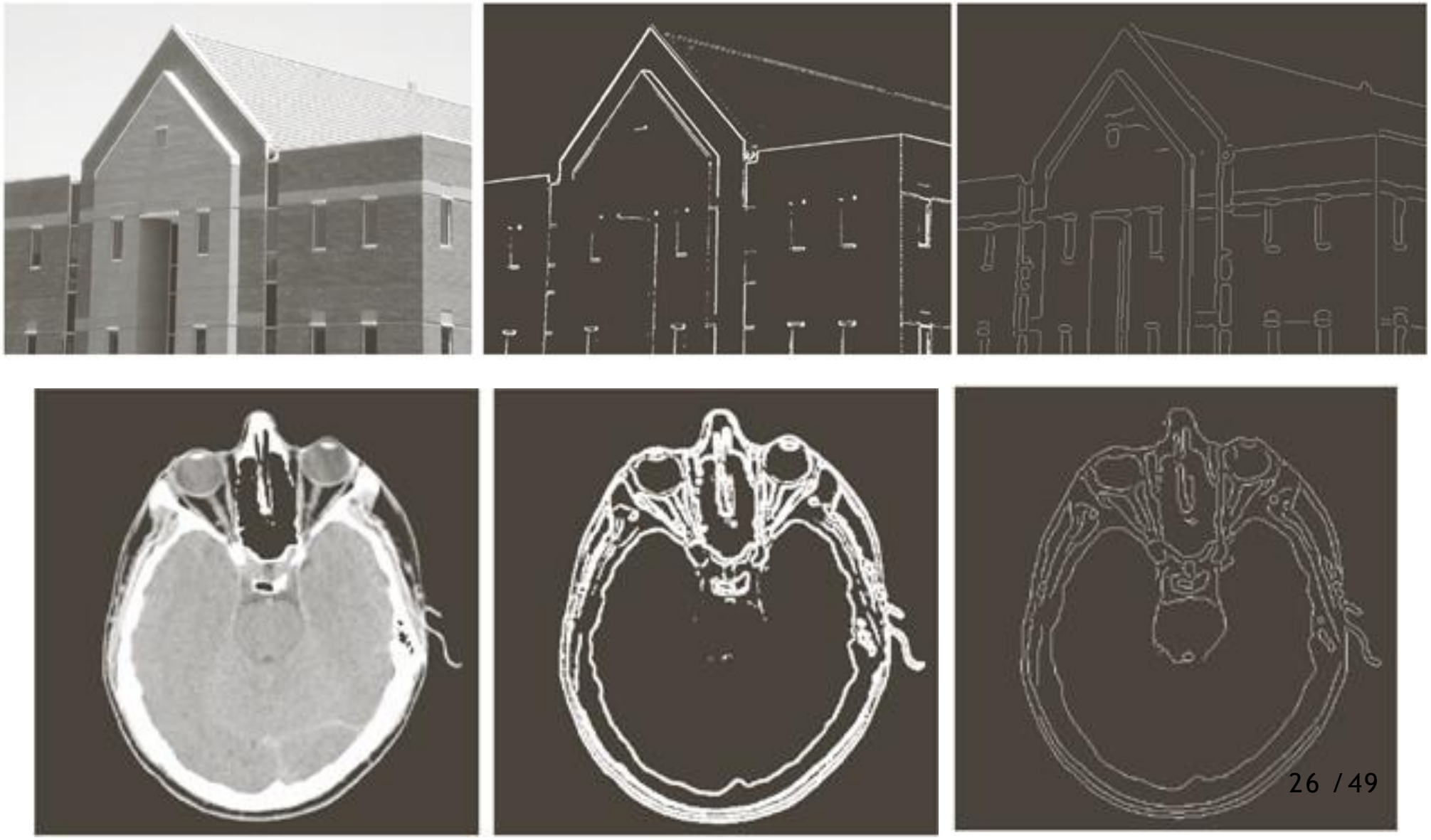
- What happens if the image is **noisy**?
- Both first and second derivatives are sensitive to noise; second derivative is more sensitive!
- Example, an 8-bit image with additive Gaussian noise of zero mean and standard deviation of **0.0**, **0.1**, **1.0** and **10** intensity levels.
- To eliminate effects of noise, it is common to smooth the image first before edge detection.



Canny Edge Detector

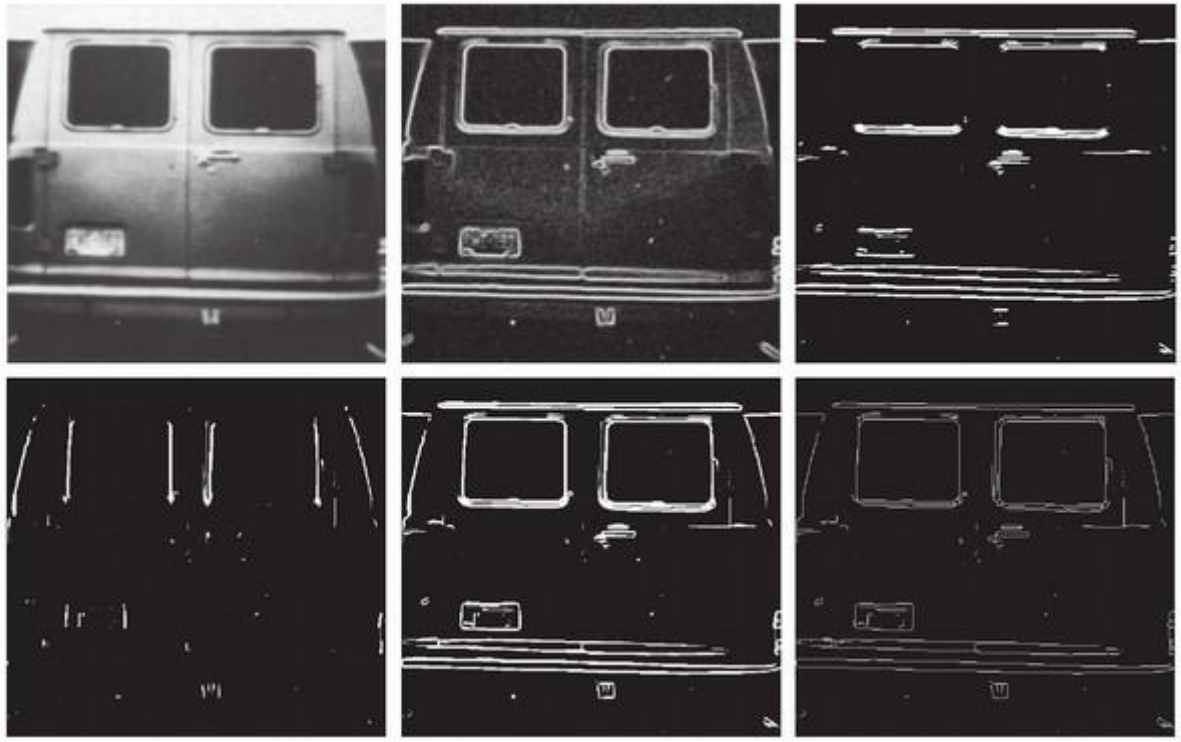
- The Canny edge detector is based on three objectives:
  - Low error rate.
  - Edge points should be well localized.
  - Single edge point response.
- Canny edge detector steps:
  1) Smooth the input image using a Gaussian filter;
  2) Compute the gradient magnitude and direction;
  3) Apply non-maximal suppression to the gradient magnitude;
  4) Use double-thresholding and connectivity analysis to detect and link edges.

Canny Edge Detector - Example



Edge Linking – Local Method

- The result of edge detection is affected by noise, breaks in the edges caused by non-uniform illumination and other effects.
- Edge detection is commonly followed by edge linking.
- In local edge linking, the gap between two edge points is filled if they have similar gradient magnitude and direction.



a b c
d e f

FIGURE 10.27
(a) Image of the rear of a vehicle.
(b) Gradient magnitude image.
(c) Horizontally connected edge pixels.
(d) Vertically connected edge pixels.
(e) The logical OR of (c) and (d).
(f) Final result, using morphological thinning. (Original image courtesy of Perceptics Corporation.)

Image Thresholding

- Given a threshold **T**, we want to partition the image into two regions: foreground (object) and background.

$$g(x,y)=\begin{cases}1 & \text{if } f(x,y)>T\\ 0 & \text{if } f(x,y)\leq T\end{cases}$$

- When **T** is a constant applied to the entire image, it is called ***global thresholding***. When the value of **T** changes over the image, it is called ***variable thresholding***.
- When the value of **T** depends on the spatial coordinate **(x,y)**, then it is called ***dynamic*** or ***adaptive thresholding***.
- We can have multiple thresholds for an image:

$$g(x,y)=\begin{cases}\text{a.} & \text{if } f(x,y)>T_2\\ \text{b.} & \text{if } T_1\in f(x,y)\leq T_2\\ \text{c.} & \text{if } f(x,y)\leq T_1\end{cases}$$

Image Thresholding

- By inspecting the histogram of an image, we can have a better idea about the optimal threshold.

$$g(x,y)=\begin{cases}1 & \text{if } f(x,y)>T\\ 0 & \text{if } f(x,y)\le T\end{cases}$$

$$g(x,y)=\begin{cases}\text{a.} & \text{if } f(x,y)>T_2\\ \text{b.} & \text{if } T_1\in f(x,y)\le T_2\\ c & \text{if } f(x,y)\le T_1\end{cases}$$

a b

FIGURE 10.32 Intensity histograms that can be partitioned (a) by a single threshold, and (b) by dual thresholds.

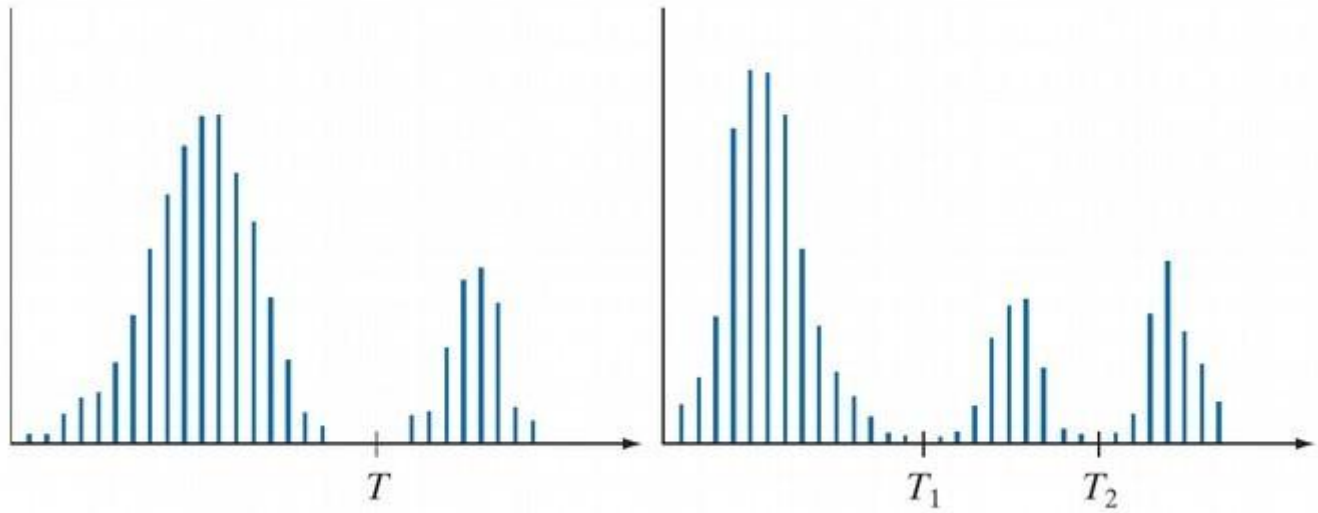
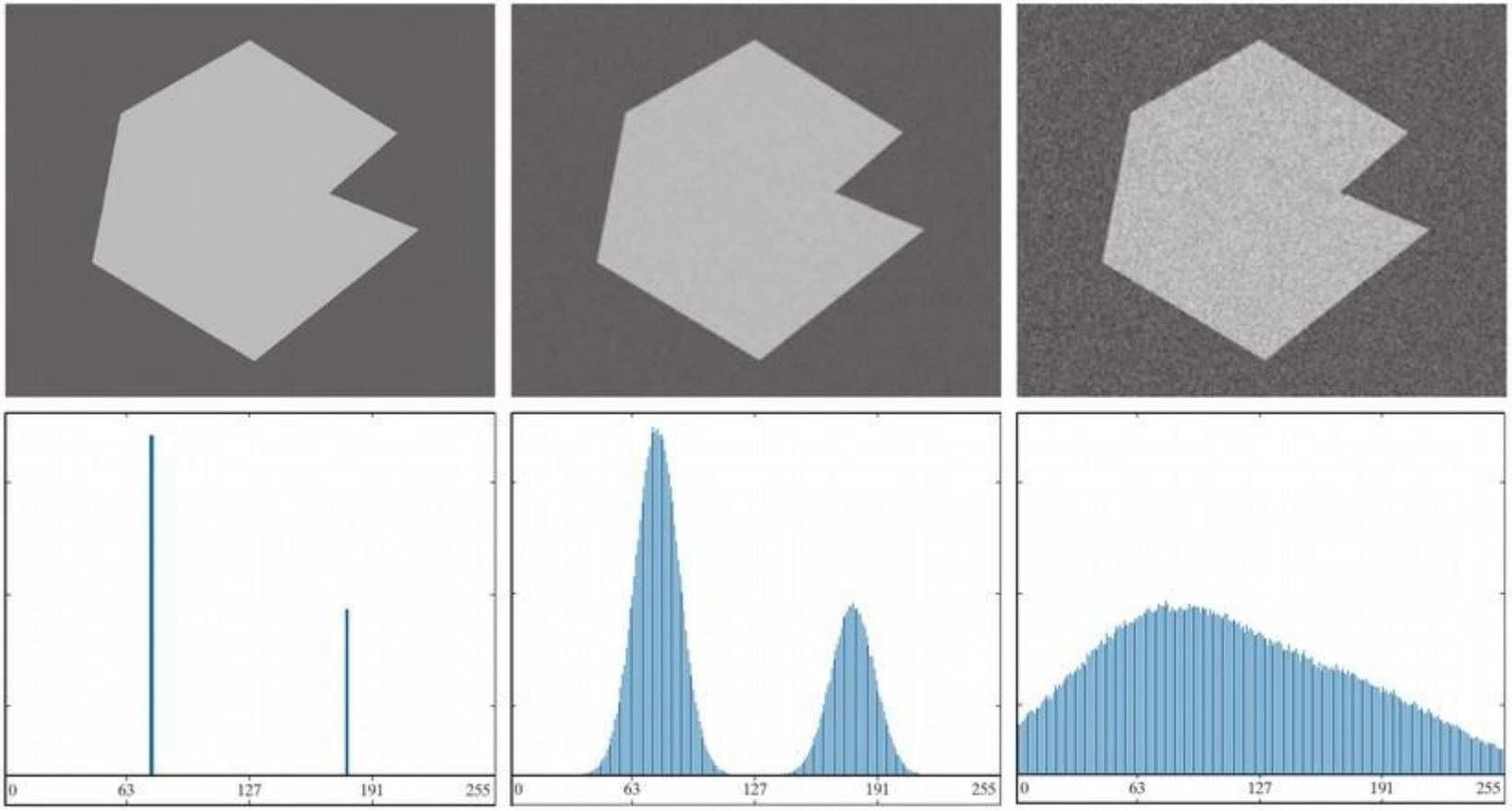


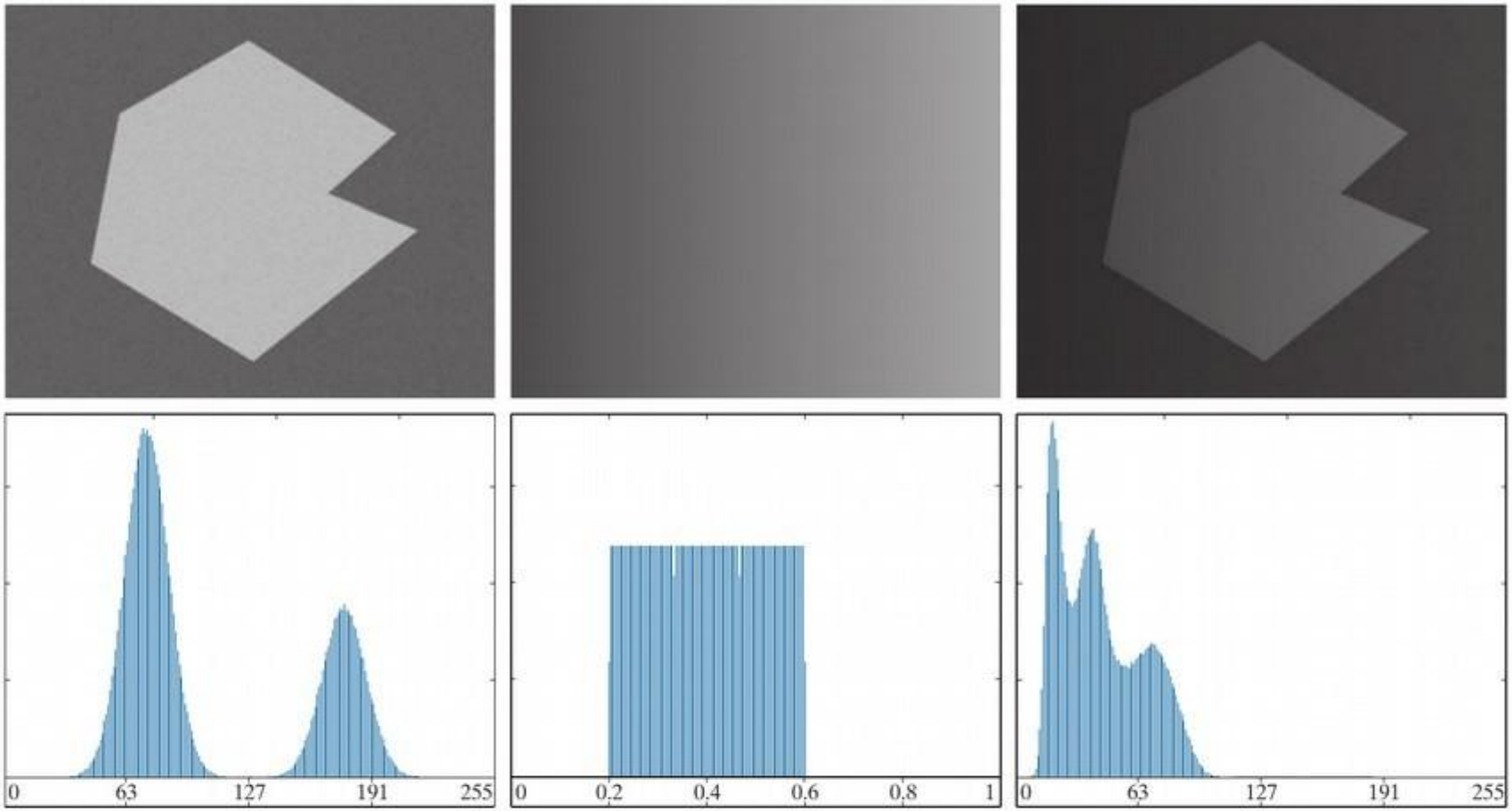
Image Thresholding - Noise



a b c
d e f

FIGURE 10.33 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d) through (f) Corresponding histograms.

Image Thresholding – Illumination and Reflectance



a b c
d e f

FIGURE 10.34 (a) Noisy image. (b) Intensity ramp in the range [0.2, 0.6]. (c) Product of (a) and (b). (d) through (f) Corresponding histograms.

Global Image Thresholding

- When the intensity distributions of objects and background pixels are sufficiently distinct, it is possible to use a single threshold for the whole image.
- The steps for global image thresholding are as follows:

  1) Select an initial **estimate** for the global threshold **T**;

  2) Segment the image using **T**; This produces two groups of pixels: $\mathbf{G_1}$ consisting of pixels with intensity values **greater** than **T**, and $\mathbf{G_2}$ consisting of pixels with intensity values **less** than or **equal** to **T**.

  3) Compute the average intensity values $\mathbf{m_1}$ and $\mathbf{m_2}$ for the pixels in $\mathbf{G_1}$ and $\mathbf{G_2}$, respectively.

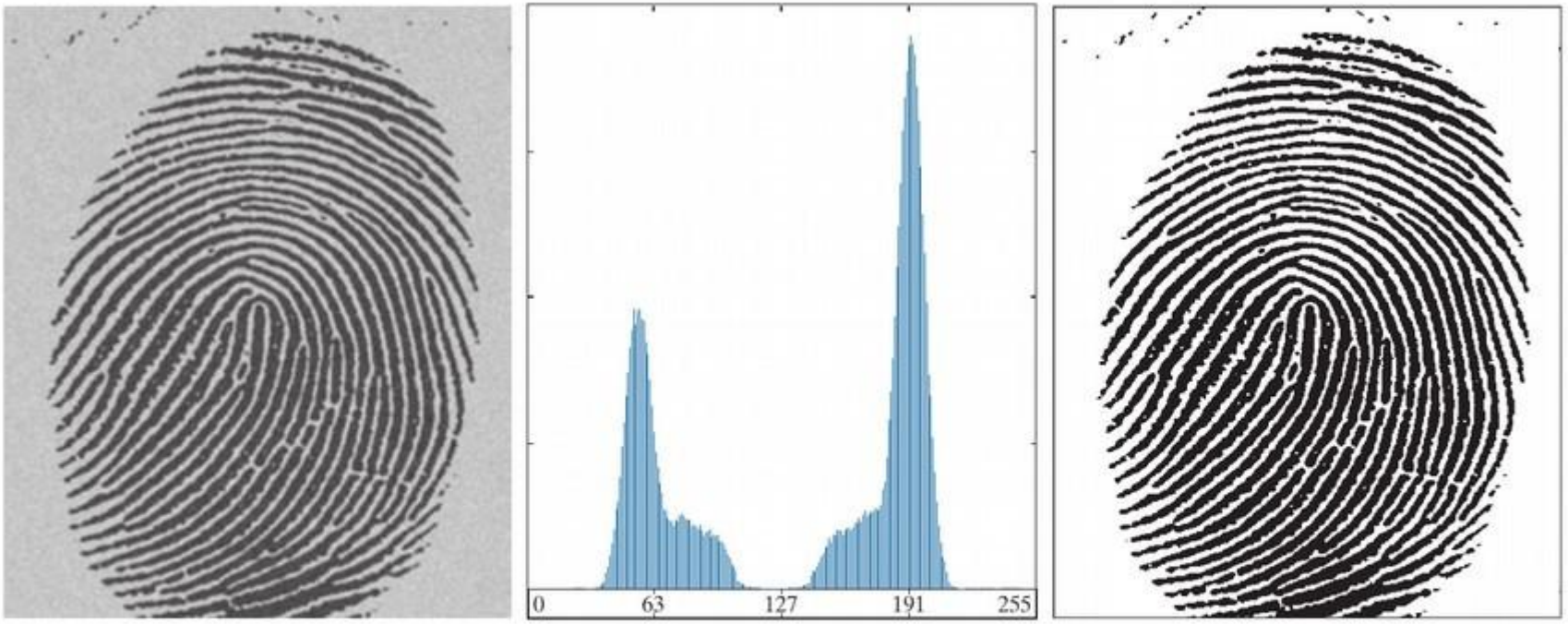
  4) Compute a new threshold value midway between $\mathbf{m_1}$ and $\mathbf{m_2}$:

$$T = \frac{1}{2}(m_1 + m_2)$$

  5) Repeat steps **2** to **4** until the difference between values of **T** in successive iterations is smaller than a predefined **ΔT**.

Global Image Thresholding - Example

- The initial threshold is set as the average intensity of the image.
- After three iterations, the global threshold **T=125**.



a b c

FIGURE 10.35 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (thin image border added for clarity). (Original image courtesy of the National Institute of Standards and Technology.).

Optimum Global Thresholding – Otsu's Method

- Thresholding can be considered as a **statistical-decision** theory problem.
- The objective is to minimize the average error in assigning pixels to two or more groups or classes.
- **Otsu's** method tries to maximize the ***between-class variance*** which measures the statistical discrimination between the classes.
- This is equivalent to minimizing the ***in-class variance*** for each group or class of pixels.
- The derivation of the equations is based on **normalized** histograms.
- Remember, the normalized histogram of an image of size **MxN** with intensity values **i=0,1,2,…, L-1**:

$$p_i = \frac{n_i}{MN}$$

$$\sum_{i=0}^{L-1} p_i = 1 \qquad p_i \geq 0$$

Optimum Global Thresholding – Otsu's Method

- Assume we select a threshold **T(k)=k** for **0<k<L-1** to partition the input image into two classes $\mathbf{c_1}$ and $\mathbf{c_2}$.
- $\mathbf{c_1}$ consists of pixels with intensity values in the range **[0, k]**; $\mathbf{c_2}$ consists of pixels with intensity values in the range **[k+1, L-1]**.
- The ***probability*** of each class can be calculated by:

$$P_1(k) = \sum_{i=0}^{k} p_i \qquad P_2(k) = \sum_{i=k+1}^{L-1} p_i = 1 - P_1(k)$$

- The ***mean intensity*** value of each class is calculated as:

$$m_1(k) = \frac{1}{P_1(k)} \sum_{i=0}^{k} i\, p_i \qquad m_2(k) = \frac{1}{P_2(k)} \sum_{i=k+1}^{L-1} i\, p_i$$

Optimum Global Thresholding – Otsu's Method

- We can define the ***cumulative mean*** (average intensity) up to level **k** as:

$$m(k) = \sum_{i=0}^{k} i\, p_i$$

- Also the ***average intensity*** of the entire image is given by:

$$m_G = \sum_{i=0}^{L-1} i\, p_i$$

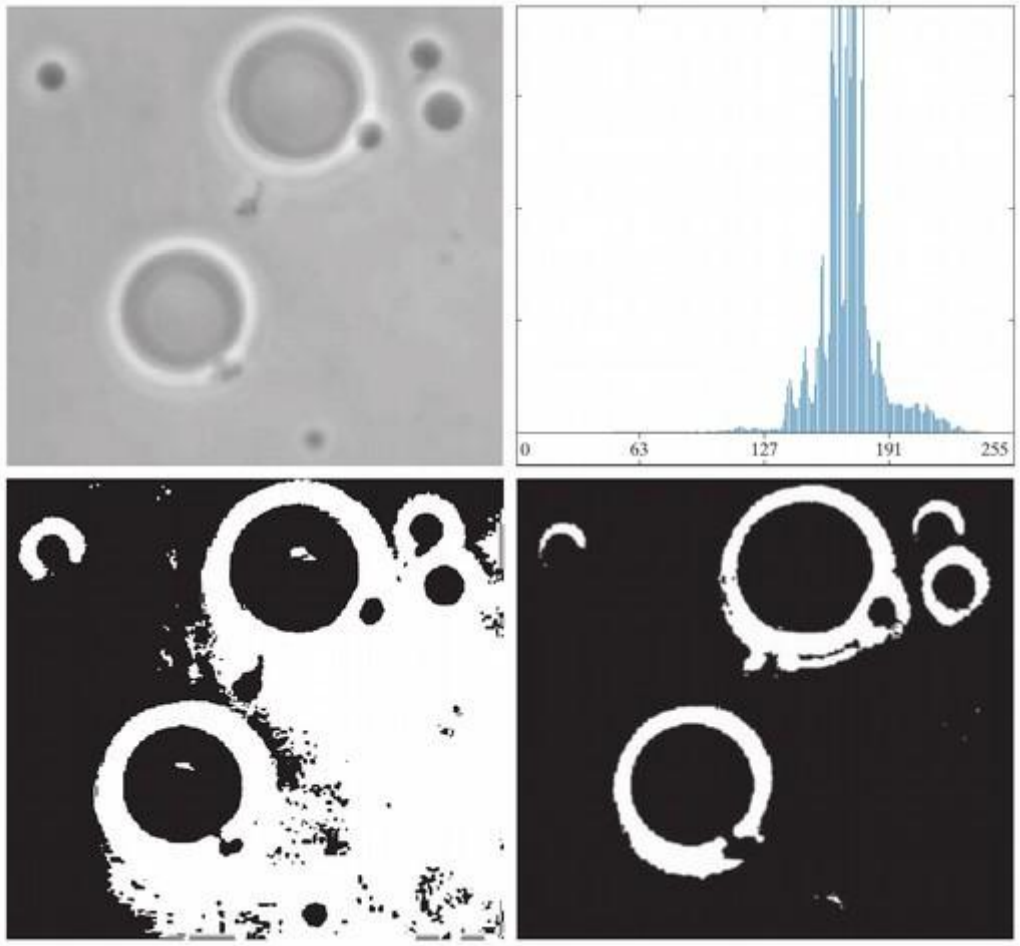
- Based the previous equations, it is obvious that:

$$P_1 m_1 + P_2 m_2 = m_G$$

$$P_1 + P_2 = 1$$

Otsu's Method - Example

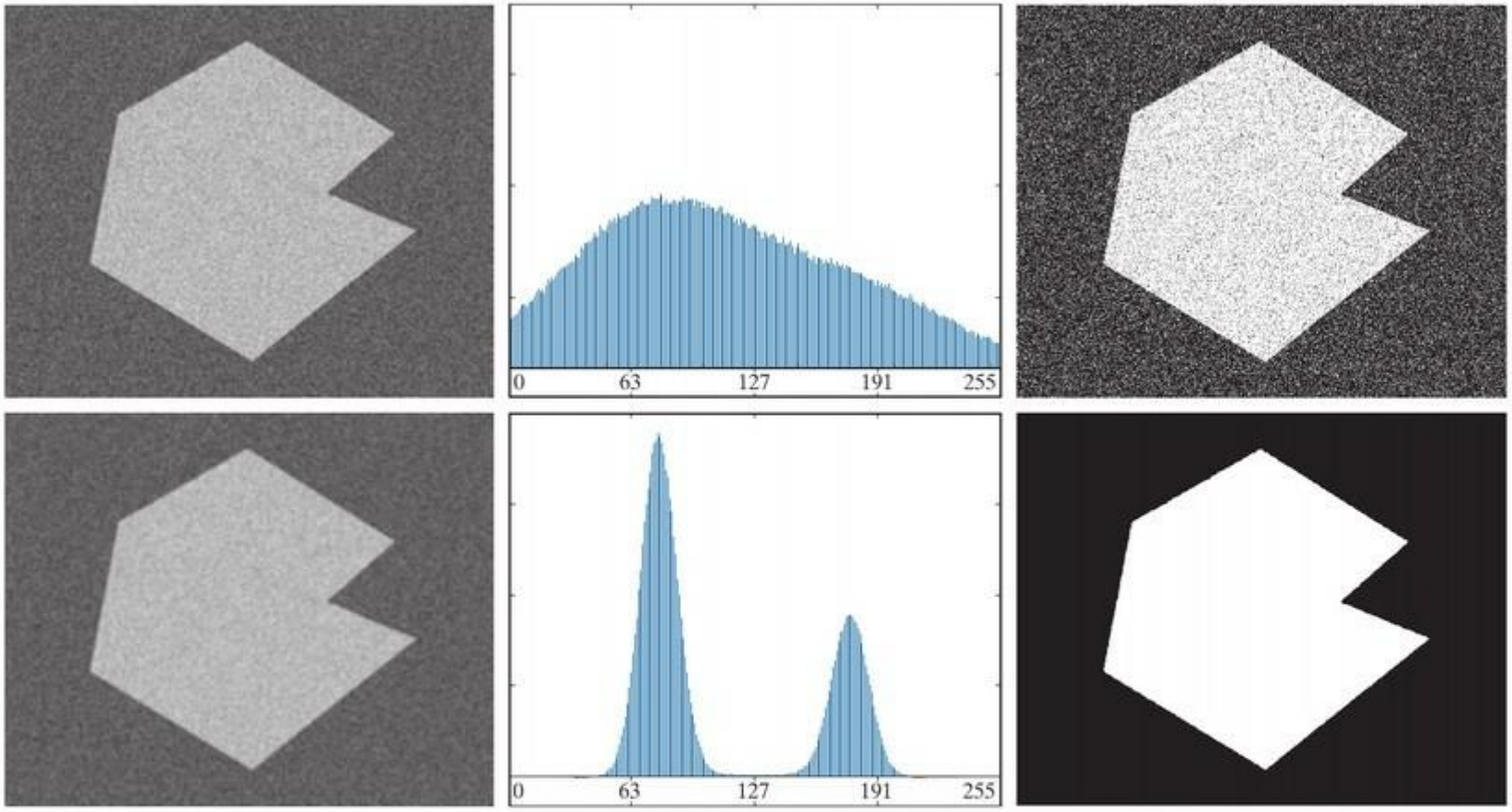
- Global method's threshold is **169**; Otsu's method's threshold is **182**.



a b
c d

FIGURE 10.36 (a) Original image. (b) Histogram (high peaks were clipped to highlight details in the lower values). (c) Segmentation result using the basic global algorithm from Section 10.3. (d) Result using Otsu's method. (Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)

Otsu's Method – Example With Noisy Image

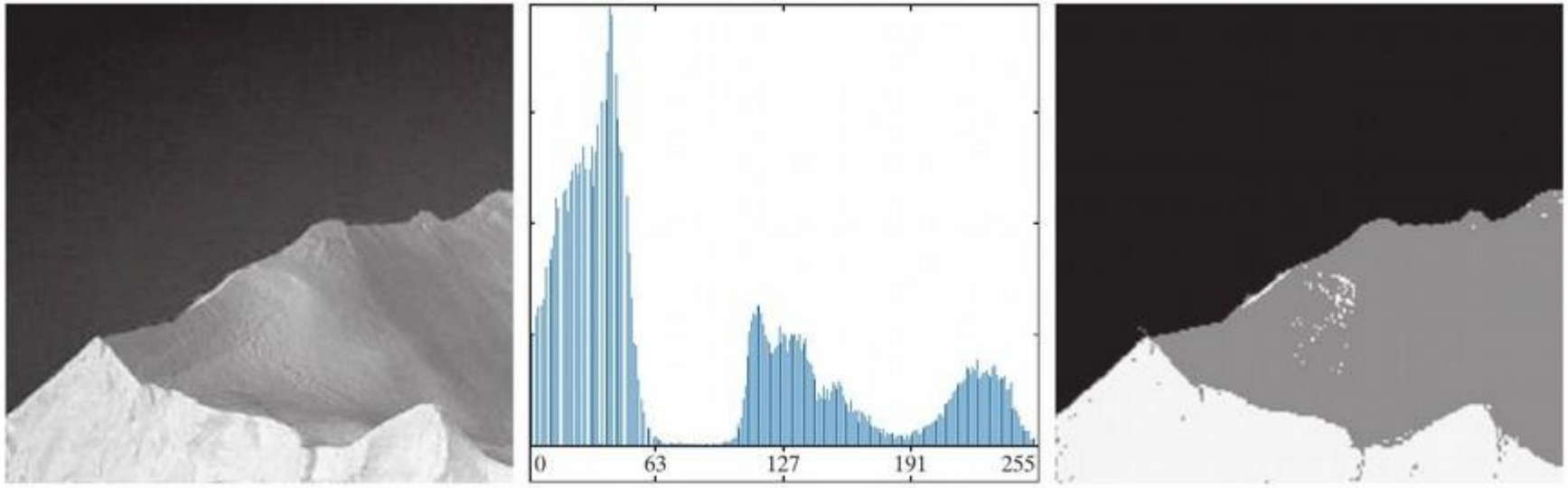


a b c
d e f

FIGURE 10.37 (a) Noisy image from Fig. 10.33(c) and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging kernel and (e) its histogram. (f) Result of thresholding using Otsu's method.

Multiple Thresholds - Example

- Otsu's method can be extended to generate more than one threshold.



a b c

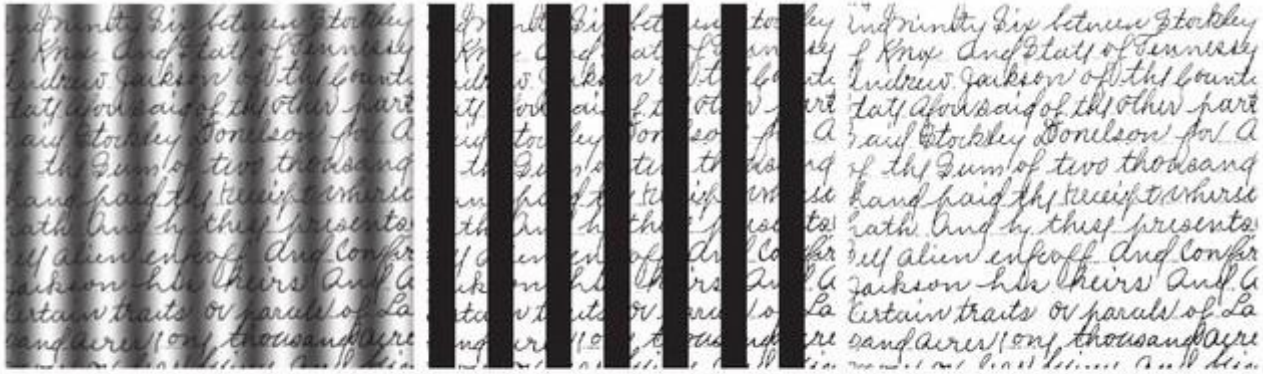
FIGURE 10.42 (a) Image of an iceberg. (b) Histogram. (c) Image segmented into three regions using dual Otsu thresholds. (Original image courtesy of NOAA.)

Variable Thresholding Using Moving Averages

- The thresholding is done line by line to reduce illumination bias.

a b c

FIGURE 10.44 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.



a b c

FIGURE 10.45 (a) Text image corrupted by sinusoidal shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages..